6	**(a)**	**(i)**	ke^{kx}	**B1** **[1]**	**1.2**		
6	**(a)**	**(ii)**	Reason referring to growth proportional to profit.	**E1** **[1]**	**3.3**	E.g. As more people hear about the business, they will sell more so it is reasonable that the rate of growth is proportional to the profits	

Question			Answer	Marks	AOs		Guidance
6	(b)		$ln\, y = ln\, A + kx$ Equation is of form "$y = mx + c$" so a straight line - hence model is consistent with graph	M1 E1 [2]	1.1 2.4		
6	(c)		$ln\, A = 1.9$ $A = 6.686$ $k = \frac{0.4}{1.6}$ $k = 0.24$ to 0.25	M1 A1 M1 A1 [4]	1.1a 2.2a 1.1 2.2a	Intercept = 1.9 not enough for M1 One or more d.p. Attempt to find gradient	Allow $e^{1.9}$ $\frac{11}{45}$ gets M1A1
6	(d)		$x = 11$ $y = 6.686 \times e^{0.25x}$ So $y = 6.686 \times e^{0.25\times 11}$ 104.586… £104 587	B1 M1 A1 [3]	3.3 3.4 3.2a	Use of model with *their A* and *k* Translation into pounds (may be to nearest thousand) (FT their *k* and *a*)	
6	(e)		Not reliable. Extrapolation may not be valid o.e. OR Fit of model is good so far and it's only two more years o.e.	E1 [1]	3.5b	E.g. • The relationship between sales and time may change • There is a limit to the market for revision resources • Changes in the economy may affect the business	'Reliable' plus 'extrapolation' does not score